AP PRECALCULUS

Unit 1 • Lesson 1.9 practice • Answer key

Lesson 1.9 • Rational Functions and Vertical Asymptotes

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Which line is a vertical asymptote of $r(x) = \dfrac{x+2}{x^2 - x - 6}$?

(A) $x = -2$

(B) $x = 3$

(C) $x = -2$ and $x = 3$

(D) $x = 6$

Solution

Answer: (B). $\dfrac{x+2}{(x-3)(x+2)}$: the factor $(x+2)$ cancels (a hole at $x = -2$), so only $x = 3$ is an asymptote.

Question 2

NO CALCULATOR

$\lim\limits_{x \to 1^+} \dfrac{2x}{x-1}$ is

(A) $+\infty$

(B) $-\infty$

(C) 2

(D) 0

Solution

Answer: (A). The numerator approaches $2 > 0$ and the denominator approaches 0 through positive values.

Question 3

NO CALCULATOR

$\lim\limits_{x \to -2^-} \dfrac{1}{(x+2)^2}$ is

(A) $-\infty$

(B) 0

(C) $\dfrac{1}{4}$

(D) $+\infty$

Solution
Answer: (D). The denominator is a square, positive on both sides of -2, and approaches 0.

Question 4

NO CALCULATOR

Which function has a vertical asymptote at $x = 4$?

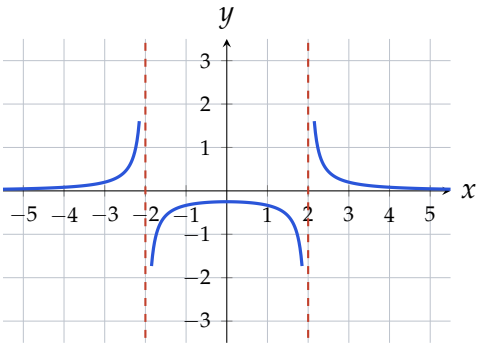
(A) $\dfrac{x-4}{x+4}$
(B) $\dfrac{x^2-16}{x-4}$
(C) $\dfrac{x}{(x-4)^2}$
(D) $\dfrac{(x-4)^2}{x}$

Solution
Answer: (C). $x = 4$ is a zero of the denominator only. In (B) the factor cancels, giving a hole.

Question 5 (graph)

NO CALCULATOR

The graph of $r(x) = \dfrac{1}{x^2-4}$ is shown, with vertical asymptotes $x = -2$ and $x = 2$. Which statement is true?



(A) $\lim_{x\to 2^-} r(x) = +\infty$
(B) $\lim_{x\to 2^+} r(x) = -\infty$
(C) $\lim_{x\to -2^+} r(x) = +\infty$
(D) $\lim_{x\to 2^-} r(x) = -\infty$

Solution
Answer: (D). Between the asymptotes the graph goes down without bound at both ends, so $\lim_{x\to 2^-} r(x) = -\infty$ and $\lim_{x\to -2^+} r(x) = -\infty$. Outside them it goes up: $\lim_{x\to 2^+} r(x) = +\infty$.

Question 6 (Free response)

NO CALCULATOR

Let $r(x) = \dfrac{x-1}{(x+2)(x-3)}$.

(a) Find the vertical asymptotes of the graph of r.
(b) For each vertical asymptote $x = a$, find $\lim\limits_{x\to a^-} r(x)$ and $\lim\limits_{x\to a^+} r(x)$. Show how you determine the signs.

Solution
(a) No factor cancels, so $x = -2$ and $x = 3$.

(b) Near $x = -2$ the numerator is close to -3 and $(x-3)$ is close to -5. For $x < -2$, $(x+2) < 0$, so the denominator is positive and $r < 0$: $\lim_{x \to -2^-} r(x) = -\infty$. For $x > -2$ the denominator is negative and $r > 0$: $\lim_{x \to -2^+} r(x) = +\infty$.
Near $x = 3$ the numerator is close to 2 and $(x+2)$ to 5. For $x < 3$ the denominator is negative: $\lim_{x \to 3^-} r(x) = -\infty$. For $x > 3$ it is positive: $\lim_{x \to 3^+} r(x) = +\infty$.